Gayatri Vidya Parishad College of Engineering Autonomous

Electronics and Communication Engineering | Course Code: EC202

Analog Electronic Circuits - Analog Circuits Small-Signal & Op-Amp Analysis

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Overview: Hybrid-pi model, Voltage-series and current-shunt feedback, Barkhausen criterion for

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Small-Signal Transistor Modeling
BJT hybrid-pi model, Common Emitter, Common Collector configurations, High frequency response, Mille

Unit 2: Feedback Amplifiers & Stability
Four basic feedback topologies, Desensitivity factor, Gain-bandwidth product, Nyquist stability crit

Unit 3: Sinusoidal Oscillators
Barkhausen criterion, RC phase-shift oscillator, Wien bridge oscillator, Hartley, Colpitts, and Crys

Unit 4: Operational Amplifiers & Active Filters
Ideal vs Practical op-amp specs, CMRR, Slew rate, Inverting/Non-inverting configurations, First and

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive the voltage gain, input impedance, and output impedance of a CE amplifier using the low-frequen
2. State Barkhausen criterion and derive the frequency of oscillation and minimum amplifier gain for a Wi
3. Design a second-order low-pass Sallen-Key active filter with a cutoff frequency of 5 kHz and Butterwor

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Adel S. Sedra, Kenneth C. Smith - Microelectronic Circuits Oxford University Press 7th Ed
* Jacob Millman, Christos C. Halkias - Integrated Electronics McGraw-Hill

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